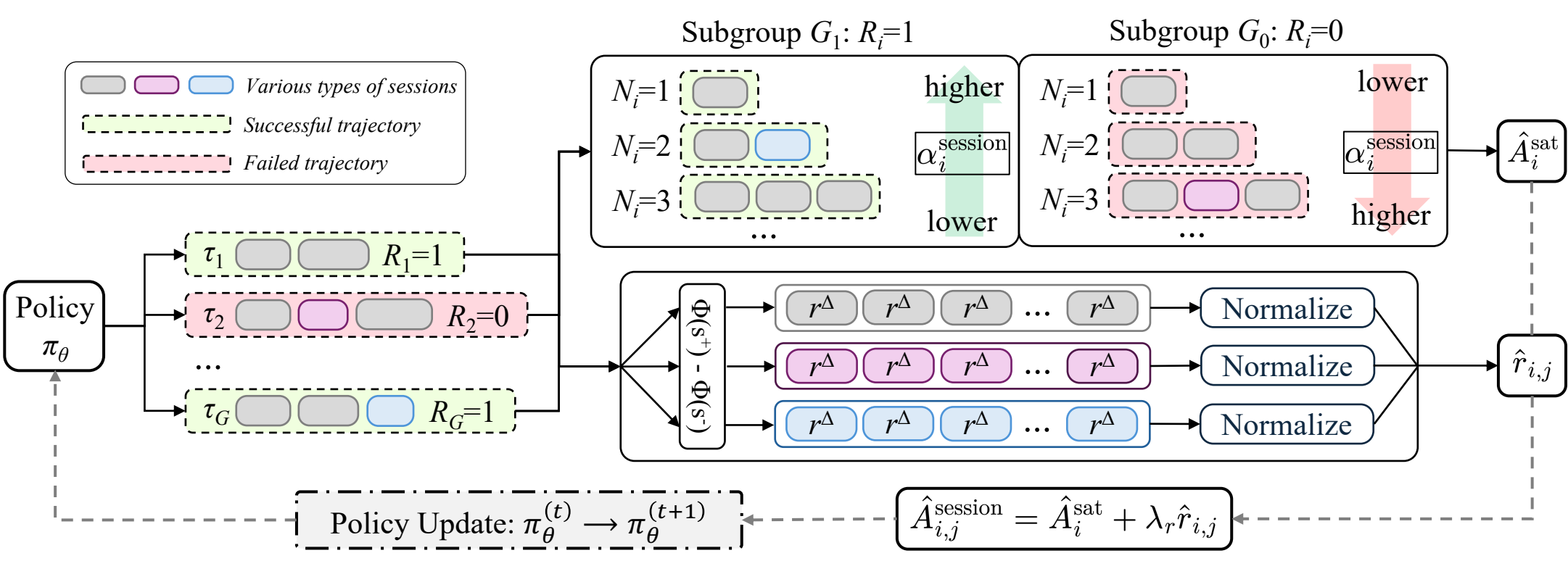

Various types of sessions
Successful trajectory
Failed trajectory
Subgroup G_1: R_i=1
Subgroup G_0: R_i=0
N_i=1
N_i=2
N_i=3
...
higher
lower
$\alpha_i^{\text{session}}$
$\hat{A}_i^{\text{sat}}$
Policy π_θ
τ_1 R_1=1
τ_2 R_2=0
τ_G R_G=1
$\Phi(s_+)$ - $\Phi(s_-)$
r^Δ
Normalize
$\hat{r}_{i,j}$
Policy Update: $\pi_\theta^{(t)} \longrightarrow \pi_\theta^{(t+1)}$
$\hat{A}_{i,j}^{\text{session}} = \hat{A}_i^{\text{sat}} + \lambda_r \hat{r}_{i,j}$